05 Owasp

OWASP Top 10 Assessment — MPPT_ALGORITHMS

Company: Anticloud FZ LLE **Standard:** OWASP Top 10 2021

#	Category	Mitigation	Status
A01	Broken Access Control	Zero-trust mTLS: every call authenticated	PASS
A02	Cryptographic Failures	AES-256-GCM at rest, TLS 1.3 in transit	PASS
A03	Injection	Parameterized queries, no shell exec from user input	PASS
A04	Insecure Design	Threat model documented, offline-first by design	PASS
A05	Security Misconfiguration	Single binary: no misconfigurable cloud console	PASS
A06	Vulnerable Components	SBOM embedded, deterministic builds, pinned deps	PASS
A07	Auth & Session Failures	mTLS short-lived certs, no session tokens	PASS
A08	Software & Data Integrity	AIOSS + Provenance chain on every mutation	PASS
A09	Security Logging	Local observability: all events logged locally	PASS
A10	SSRF	No outbound HTTP in offline mode, allowlist only	PASS

Additional: LLM Top 10 (OWASP LLM01-LLM10 2025)

#	Category	Mitigation	Status
LLM01	Prompt Injection	Input sanitization + output validation	PASS
LLM02	Insecure Output Handling	Structured output validator, schema enforcement	PASS
LLM03	Training Data Poisoning	Model weights verified via hash, not retrained online	PASS
LLM04	Model Denial of Service	Token budget enforced, request queuing	PASS
LLM05	Supply Chain Vulnerabilities	SBOM, deterministic build, pinned GGUF hash	PASS
LLM06	Sensitive Info Disclosure	Sovereign Memory encrypted, no plaintext logs of prompts	PASS
LLM07	Insecure Plugin Design	No plugins; all tools compiled into single binary	PASS
LLM08	Excessive Agency	PAX operates in read-only mode by default	PASS
LLM09	Overreliance	Human-in-the-loop required for all write operations	PASS
LLM10	Model Theft	Model runs locally; GGUF never transmitted externally	PASS